

Automatic detection of crop rows based on multi-ROIs



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ABSTRACT

This paper presents a new automatic and robust crop rows detection algorithm for a vision-based agricultural machinery guidance system. The algorithm consists of five steps: gray-level transformation, binarization, candidate center points estimation, real center points confirmation and crop rows detection. During the procedure of candidate points estimation, a multi-ROIs is constructed, which integrates the features of multi-rows based on a geometry constraint that inter-row space is approximately equal into a single processing of optimization. This strategy ensures the center points of crop rows could be detected in a high weedy field or in a large barren patch of the field. For the fourth step, we used the clustering method to confirm the real center points indicating crop rows. And lastly, the linear regression method was employed to fit the crop rows. Three image data sets, taken from a wheat, corn and soybean under different natural and field conditions, were used to evaluate the detection rate, detection accuracy and processing time of the algorithm. Thorough an experimental comparison with the standard Hough transform (SHT), it demonstrates the proposed method outperforms SHT either in detection rate, detection accuracy or computation time. The algorithm requires about 61 ms to recognize crop rows for a 640×480 pixels image while the detection rate reaches 93%, detection accuracy reaches 0.0023°.

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1. Introduction

With the development of precision agriculture, the autonomous navigation of agricultural machinery has received more and more attention. It has a wide range of applications, such as automatic spraying pesticide, weeding, harvesting and tilling. The potential advantages of autonomous navigation vehicles may be summarized as the following aspects: (1) reduce operator fatigue and improve vehicle positioning accuracy (Gerrish, Fehr, Van Ee, & Welch, 1997); (2) minimize compaction, leading to improved yield and profitability (Gan-Mor and Clark, 2001); (3) improve efficiency; and (4) enhance operation safety (Han, Zhang, Ni, & Reid, 2004). According to a literature review study (Billingsley & Schoenfisch, 1995; Hague, Marchant, & Tillett, 2000; Li, Imou, Wakabayashi, & Yokoyama, 2009; Mousazadeh, 2013; Wilson, 2000), various guidance technologies, including mechanical guidance, optical guidance, radio guidance, Global Positioning System (GPS) and machine vision navigation, etc. have been developed. Especially what deserves to be mentioned is that the vision navigation has already become more and more popular in the agricultural domain, it will largely contribute to the availability of low cost cameras and the plethora of computer vision techniques that could

be readily applied (Hiremath, Van Der Heijden, Van Evert, Stein, & Ter Braak, 2014).

Most crops are cultivated in rows, so, for most agricultural operations such as mentioned above, an important step for an agricultural vehicle is the development of a crop row detection system (Keicher & Seufert, 2000), which supply enough navigation information to ensures it accurately follow a row of crops while moving forwards.

1.1. Literature review

A considerable number of studies have been conducted to develop and apply techniques for detecting crop rows with image processing techniques (Zhang, 2011). Typically, existing approaches can mainly fall into the following categories according to their detection principle:

- (1) Methods based on Hough transform (HT): Hough transform (Hough, 1962) is a feature extraction technique in digital image processing, image analysis and computer vision, and has long been widely used to detect geometric features such as lines and curves (Duda & Hart, 1972). The basic idea involved in locating lines with the HT is point-line duality. With the coordinate transformation, the collinear points in image space correspond to concurrent lines in parameter space by voting. As a result, the problem of line detection

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in image space can be convert the problem of point detection in the parameter space, which can be finished by accumulating the votes and detecting the peaks in the parameter space. To detect the peaks, we usually use an accumulator with finite resolution to quantify the parameter space into cells. Therefore, the accumulation on the cells enables us to detect peaks of the voting and then get the parameters of lines. In agricultural engineering field, the HT and some variations on Hough's original transform have been extensively used for crop row detection in vision-based agricultural navigation systems (Ericson & Åstrand, 2009; Gée, Bossu, Jones, & Truchetet, 2008; Marchant, 1996; Marchant & Brivot, 1995; Reid & Searcy, 1986; Torii, Kitani, Okamoto, Kusano, & Inou, 1995; Van Evert et al., 2006; Åstrand & Baerveldt, 2005). Rovira-Más, Zhang, Reid, and Will (2005) used the Hough transform-based path-finding method to detect crop rows. Bakker et al. (2008) performed several steps for image preprocessing, and lastly sugar beet rows were detected applying gray scale Hough transform (Lo & Tsai, 1995). Despite its robust against noise, the computational burden associated to its voting scheme has prevented its agricultural applications to achieve real-time performances, except for very small images (Fernandes & Oliveira, 2008). In order to speed up the calculation, Xu, Oja, and Kultanen (1990) proposed a randomized Hough Transform (RHT) algorithm. The core concept of the RHT is randomly selects two edge pixels each time with equal probability in the image space and then their corresponding mapped point in the parameter space is accumulated by voting in a accumulator (Xu et al., 1990). The procedure is continued until some cells in the accumulator larger than a given threshold, at this moment, each of them represents a straight line. The RHT was applied by Ronghua and Lijun (2011). However, the method will be unfeasible in the following two cases: (a) when an image with high weed pressure, the randomly-selected paired points in image usually derived from weed, which cause the detection lines are incorrect. (b) when an image with serious crop loss, some cells in the accumulator may be less than a given threshold, which lead some crop rows cannot be detected.

- (2) Methods based on least squares fitting: "least squares" means that the overall solution minimizes the sum of the squares of the errors made in the results of every single equation (Least Squares, 2014). In earlier research, Billingsley and Schoenfish (1995) used linear regression based on information from three row segments to detect crop row guidance information. Recently, Montalvo et al. (2012) proposed a crop row detection method in maize fields with high weeds pressure. The strategy included three steps: image segmentation, double thresholding based on Otsu and linear regression. According to their reports, they separate weed and crops was due to the double Otsu's method, which made the least squares fitting possible. The result shows it is better than classical Hough transform approach in processing time. Although their work was valuable in detecting crop rows in maize fields, there was still a prior knowledge limitation. Because their method based on some special templates, the design of templates must be built under much previous knowledge, such as number of crop rows, the location of each row in the image, etc., which would bring some troubles and inconvenience to operation. In a word, least squares fitting is highly affected by image noise. In our proposed method we already classified center points indicating crop rows according to different the value of k (see Section 2.2.4), making the least squares fitting a realistic approach.

- (3) Methods based on vanishing point: Pla, Sanchiz, Marchant, and Brivot (1997) proposed a guidance information extraction method based on vanishing point. The procedure included the image segmentation, the row skeletons extraction, the straight line extraction, the detection of vanishing point and the crop row recovery. However, this process shows a heavy reliance on the region skeletons, which are sometimes difficult to extract, especially in images with a high weed pressure (Guerrero et al., 2013). Moreover, this approach was computationally demanding (Leemans & Destain, 2006).
- (4) Methods based on stereovision: Stereovision also has been used successfully to detect crop rows. For example, Kise, Zhang, and Rovira Más (2005) used stereo images to analyze the geometry of field terrain to detect crop rows. Wang, Zhang, Rovira-Más, and Tian (2011) developed a stereo image-processing algorithm to detect and track ground features. These features were used as reference points to calculate the lateral offset. Field validation tests showed that this algorithm could provide accurate relative lateral offset measurements. However, this approach was based on the assumption of "no rotation", and this assumption will be invalid in reality, because the bumpiness of the path could induce camera rotation.
- (5) Methods based on horizontal strips: Søgaard and Olsen (2003) proposed a crop rows determination by image analysis without segmentation. For a gray-level image after transforming, they divided it into 15 horizontal strips, and then computed the vertical sum of gray values in each strip, the maximum denote the center of crop row in each strip. Lastly, crop rows are detected by the center points. Sainz-Costa, Ribeiro, Burgos-Artizzu, Guijarro, and Pajares (2011) developed a strategy based on analysis of video sequences to detect crop rows. They used the lower half part of the image to detect. The image processing experienced five steps: segmentation, morphological opening, horizontal strips dividing, vertical average, row centers extracting and crop rows finding. The above approaches can work well under low weed pressure. However, under the complex conditions such as image with high pressure or serious plant missing, the strip features are usually cannot correctly detect.

1.2. Objectives of the study

According to the above mentioned, no previous work considers exploiting the implied information based on geometric property of crop rows. So, from a new point of view, crop rows detection based on multi-ROIs is presented. The overall goal of the present study is to develop an efficient image processing algorithm on the basis of analyzing the virtue of the previous approach for recognizing crop rows.

The specific objectives are as follows:

- To develop an image extraction approach for estimating the center points indicating of crop rows, which can work under different natural field conditions.
- To develop a real-time line detection algorithm for crop rows detection.
- To evaluate the applicability of the proposed method.

1.3. Introduction to the proposed strategy

In this paper, an automatic crop rows detection method is proposed. It differs from previous work in that an implicit geometric constraint is employed, which make it possible to well find the

center points of crop rows in a high weed pressure or serious plant deficiencies even in the case of no prior knowledge. The flow diagram of the proposed method is schematically presented as Fig. 1, which includes five steps. In the first two steps, i.e., the image graying and binarization, the color indices and Otsu's method were used to obtain a good segmentation result. After this, in order to extract the candidate center points indicating the crop row centers, a multi-ROIs based on the fact that inter-row space is approximately equal in the field was constructed, which fuses the features of multi-rows. Then, to obtain the real center points of crop rows, pseudo points are removed by classification algorithm and lastly the crop rows are detected according to the least squares fitting. So, the main contribution of our work is on the one hand, we create a multi-ROIs model to improve the robustness of the approach; on the other hand, we use a classification method to make the least squares feasible, which further improve the real-time property.

1.4. Paper organization

The outline of this paper is organized as follows. Section 1 is the general introduction. Section 2 describes materials and methods used for the strategy in detail. Section 3 shows the results obtained by employing the proposed method. Lastly, the conclusions are presented in Section 4.

2. Strategy for plant identification—materials and methods

2.1. Material—image acquisition

In the process of image acquisition, a Samsung S750 color camera was mounted on a farm vehicle. The focal length of the camera lens was 5.8 mm, the aperture setting was $f/7.1$, ISO was set 80 and the exposure time in image acquisitions was changed in the range of $1/250$ – $1/60$ s according to the different illuminations. Image processing was performed using a personal computer with an Intel Core 2 Duo CPU, 2.93 GHz, 1.99 GB RAM, and the algorithm was developed using C programming language and MATLAB 7.0 (MathWorks) under a Windows XP operating system.

Sampling operations were performed at two different fields, which are located in National Research and Demonstration Base of Precision Agriculture in Xiaotangshan Town of Beijing and Farm of Chinese Academy of Agricultural Sciences in Beijing. Three kinds of agronomic images were selected in this study (wheat, corn and soybean). The inter-row space was approximately 20, 45 and 35 cm for wheat, corn and soybean, respectively. They were acquired under natural lighting in different years and seasons.

The digital images were stored as 24-bit color images with a resolution of 1024×768 pixels and saved in RGB color space with the JPEG format. After the JPEG images were obtained, it is necessary to carry out the down sampling processing to reduce the computational load. Here, the classic resampling method—bilinear interpolation method is employed due to its simplicity and effectiveness. Fig. 2 shows an example of resampling operations. After bilinear interpolation, the image size shrank from the original size of 1024×768 to 640×480 pixels.

2.2. Image processing

In order to automatic detect the center lines of crop rows, it requires the following five major steps: (1) gray-level transformation, (2) binarization, (3) the estimation of candidate center points, (4) real center points confirmation and (5) crop rows detection.

2.2.1. Gray-level transformation

The first step, which is similar to almost all image processing crop rows detection methods, is the gray-level transformation. It aims to emphasize the living plant tissue and weaken the rest of image. This can be done in an RGB color space. So, color indices (Meyer, Mehta, Kocher, Mortensen, & Samal, 1998; Woebbecke, Meyer, Von Bargen, & Mortensen, 1995) for RGB images have been designed to emphasize vegetation pixels while restrain other elements such as soil, stones, and shadow. The famous color indices are as follows:

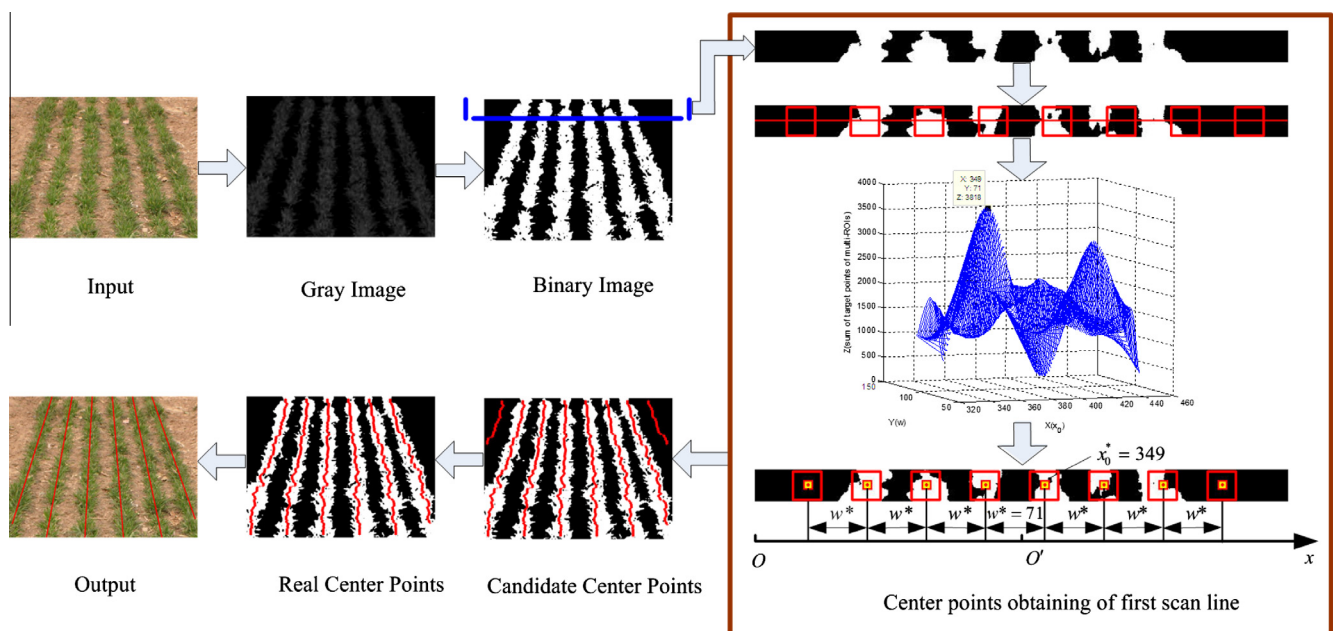


Fig. 1. Flow diagram of the automatic crop rows detection method, which consists of five main steps: (1) gray-level transformation, (2) binarization, (3) the estimation of candidate center points, (4) real center points confirmation and (5) crop rows detection.

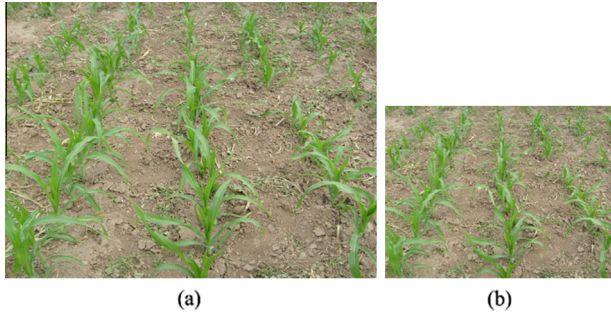


Fig. 2. The process of image down-sampling. (a) Original image, (b) the same image as (a) but using bilinear interpolation.

- (1) $2G - R - B$ (Sogaard & Olsen, 2003)
where R , G and B are the actual pixel values from the agro-
nomic images based on RGB color space.
- (2) Excess green (ExG) index (Woebbecke et al., 1995)

$$\text{ExG} = 2g - r - b \quad (1)$$

where r , g and b are the chromatic coordinates.

$$\begin{aligned} r &= \frac{R^*}{R^* + G^* + B^*}, \quad g = \frac{G^*}{R^* + G^* + B^*}, \\ b &= \frac{B^*}{R^* + G^* + B^*}, \quad r + g + b = 1. \end{aligned} \quad (2)$$

and R^* , G^* and B^* are the normalized RGB values ranging from 0 to 1 and are defined as follows:

$$R^* = \frac{R}{R_{\max}}, \quad G^* = \frac{G}{G_{\max}}, \quad \text{and} \quad B^* = \frac{B}{B_{\max}} \quad (3)$$

where $R_{\max} = G_{\max} = B_{\max} = 255$ for 24-bit color images.

- (3) $1.262g - 0.884r - 0.311b$ (Burgos-Artizzu, Ribeiro, Guijarro, & Pajares, 2011)

In the above three color indices, the second and third color indices used the normalization of the values of each spectral band, so they appeared to be insensitive to illumination intensity. In addition, the third combination was obtained by genetic algorithm optimization and was proved to be better than the ExG. In view of these facts, the color indices $1.262g - 0.884r - 0.311b$ was applied in this study. Fig. 3 displays the transformed image with the color indices from the original images displayed in Fig. 2b.



Fig. 3. Gray-level image obtained with $1.262g - 0.884r - 0.311b$ from the image in Fig. 2b.

2.2.2. Binarization

The term “binarization” means to reduce a multi-value digital signal into a two-value binary signal (Matsui & Mita, 1990). Among the global thresholding techniques for image binarization, Otsu’s method (Otsu, 1975) is one of the best threshold selections. The basic idea of Otsu’s method is select an optimal threshold value to split grey-level histogram of an image into two parts based on the principle of maximum between-cluster variance and minimum within-cluster variance. The gray-level image in Fig. 3 was processed with Otsu’s method, and the segmentation results are shown in Fig. 4. As expected, the green plants are identified and marked as white pixels.

2.2.3. The estimation of candidate center points

Prior to the estimation of the position of crop rows, it is important to determine a number of center points indicating the centers of crop rows. It is directly related to the success or failure of the crop rows detection. In fact, the corresponding center points on two adjacent horizontal scanning lines cannot have a large deviation. In view of this, once the center points on some scan line were got, the remaining center points could be easily found based on the context information. So, here, we adopt two steps to detect the center points. First, the initial center points detection on the first horizontal scanning line and second, the remaining center points detection.

(1) Initial center points detection

The initial center points detection on the first horizontal scan line is a crucial factor for the following center points estimation. Here, a sliding region of interest (referred to as SROI below), defined as a rectangle region of an image $M \times N$ (where M and N are the height and width of an image, respectively), was designed to scan the binary image from top to bottom, and left to right. In the following analysis, let $S_{(y,x)}$ be the sum of target pixels (white pixels) for a specific ROI which centered at point (y, x) . Here, $S_{(y,x)}$ was calculated as follows:

$$S_{(y,x)} = \sum_{i=y-ROIHeight}^{y+ROIHeight} \sum_{j=x-ROIWidth}^{x+ROIWidth} f(i,j) \quad (4)$$

where, $f(i, j)$, $ROIWidth$ and $ROIHeight$ refer to the gray value of point (i, j) , the width and height of ROI, respectively. In this article, the $ROIWidth$ and $ROIHeight$ are equal to 19.

On a horizontal scan line y , it is easy to imagine that the number of target pixels in a SROI will reach a maximum when SROI covers



Fig. 4. Binary image obtained with Otsu’s method for image in Fig. 3.

the crop row region. After the scanning is complete, a curve $S_{(y)}$ (the sum of all the number of target pixels within the current SROI) can also be obtained (see Fig. 5). The mean μ and standard deviation σ of $S_{(y)}$ were obtained by Eqs. (5) and (6), respectively:

$$\mu = \frac{1}{N'} \sum_{x=ROIWidth/2+1}^{N-ROIWidth/2} S_{(y,x)} \quad (5)$$

$$\sigma = \sqrt{\frac{1}{N'} \sum_{x=ROIWidth/2+1}^{N-ROIWidth/2} (S_{(y,x)} - \mu)^2} \quad (6)$$

where, N' is the number of $S_{(y)}$ when $S_{(y,x)}$ does not equal zero. In our study, the $ROIWidth = 19$.

Fig. 5 shows a process diagram about how to obtain the center points on the current scan line. The curve above the line $S_{(y)} = T$ was regarded as feature region. x_{l1} and x_{r1} denote the start and end points of the first feature region. Similarly, x_{ln} and x_{rn} represent the start and end points of the n th feature region. In addition, $x_1 = (x_{l1} + x_{r1})/2, \dots, x_n = (x_{ln} + x_{rn})/2$, on the one hand, are the midpoints between the start and end points of the first, ..., the n th feature regions, respectively; On the other hand, they denote candidate center points indicating the center lines of the crop rows. Generally, this approach yields a good result especially in cases with fewer weeds. But how we find the value of T ? According to the normal distribution theory and practical experience, T can be defined as follows:

$$T = \mu + \sigma/2 \quad (7)$$

where, the values of μ and σ are computed according to the Eqs. (5) and (6).

We take a wheat binary image as an example to illustrate the process of SROI. As seen from the Fig. 6(a), for an image with less weed, the centers of crop rows can be detected accurately using SROI method.

However, the SROI approach will not be enough if weed patches are present in the agronomic image, as can be seen in Fig. 6(b). The main reason leading to the question is that the method simply uses the information of a single ROI. Once the number of target points in the ROI hits a certain threshold T , it will give a good indication of the start of crop region even if the ROI overlaps weed patches. So, to cope with these cases, another approach based on the multi-ROIs is proposed.

The basic concept of the algorithm can be illustrated with a diagram as Fig. 7. Here, four white strips represent four crop rows,

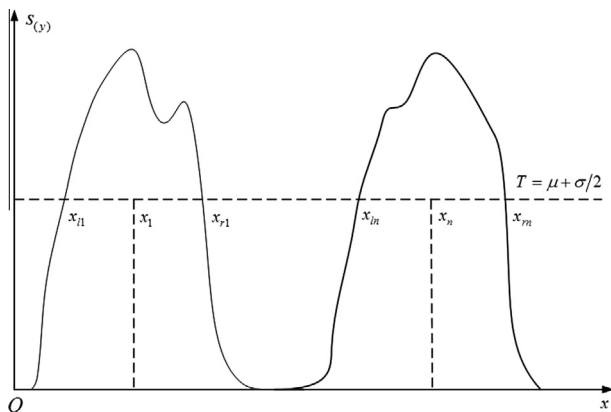


Fig. 5. Schematic diagram showing estimation of the first and n th center points (x_1 and x_n) indicating the center lines of the first and n th crop rows on the current scan line.

respectively; the remainder black region is background. We regard the current horizontal scan line as the x axis. The origin of coordinate, O , corresponds the leftmost point of the image. Point O' is the midpoint of the left and right boundary of the image. Here, points A_0 and A_1 represent the center points of crop rows from point O' to the right boundary of the image, respectively. Similarly, points B_1 and B_2 represent the center points of crop rows from point O' to the left boundary of the image, respectively.

According to the characteristics of the crop rows in a farmland, in most cases, crop are cultivated in rows and sown in a fixed model, that is to say, the inter-row space can be regarded as a constant. So, we can infer the relation as follows:

$$|B_2B_1| = |B_1A_0| = |A_0A_1| = w^* \quad (8)$$

Based on expression (8), once one of center points position and the crop rows distance on the current horizontal scan line are known, all the remaining center points can be easily calculated. Here, we assume that the coordinate value of point $A_0(x_0^*)$ (the first detected center point from point O' to the right direction) is known and the crop inter-row space w^* have been obtained, the rest center points (such as $A_1, A_2, A_3, \dots; B_1, B_2, B_3, \dots$) coordinates can be inferred by the following expression:

$$\begin{aligned} A_{kr} &= x_0^* + k_r w^* \quad (k_r = 1, 2, 3, \dots) \\ B_{kl} &= x_0^* - k_l w^* \quad (k_l = 1, 2, 3, \dots) \end{aligned} \quad (9)$$

However, the above discussion is under the condition where the values of x_0^* and w^* are known. In fact, we have no means to know their values in advance. Now the first problem to solve is how to find the values of x_0^* and w^* ? Here, a combination of several small interest regions, called multi regions of interest (multi-ROIs), is designed. It aims to find out the global optimum center points stand for the centers of crop rows.

As Fig. 7 illustrated, the multi-ROIs integrates four single ROI, which slides from top to bottom, and left to right of the image as a whole. The sum of target points (white pixels) in multi-ROIs can be obtained by Eq. (10).

$$\text{sum}(i, x_0, w) = \sum_{k=1}^n S(i, x_0 \pm kw) \quad (k = 1, 2, 3, \dots) \quad (10)$$

where, i , the coordinate value of y -coordinate, is a known value, refers to the current horizontal scan line. x_0 and w refer to the location of point A_0 and the horizontal distance between two adjacent crop rows on current scan line, respectively.

With the change of x_0 and w in a reasonable range, the $\text{sum}(i, x_0, w)$ will reach a maximum value when each ROI exactly cover the crop rows. In this moment, one notes the two optimal parameters x_0^* and w^* , which can be eventually computed according to the Eq. (11).

$$(i, x_0^*, w^*) = \arg \max_{x_1 \leq x_0 \leq x_2, w_1 \leq w \leq w_2} (\text{sum}(i, x_0, w)) \quad (11)$$

Fig. 8 shows the acquisition process of x_0^* and w^* in detail. Of course, it is only a schematic diagram, which shows an example in the middle part of the image. In actual operation, we scan the binary image from top to bottom, and left to right. That is to say, the first obtained x_0^* and w^* are from the top part of the image.

(2) Residual center points detection

Once the initial center points of crop rows were detected on a certain horizontal scan line based on the multi-ROIs, the center points of crop rows on the following scan line can be obtained by the same strategy. However, it will be a much time-consuming process for the center points detection on every scan line if we

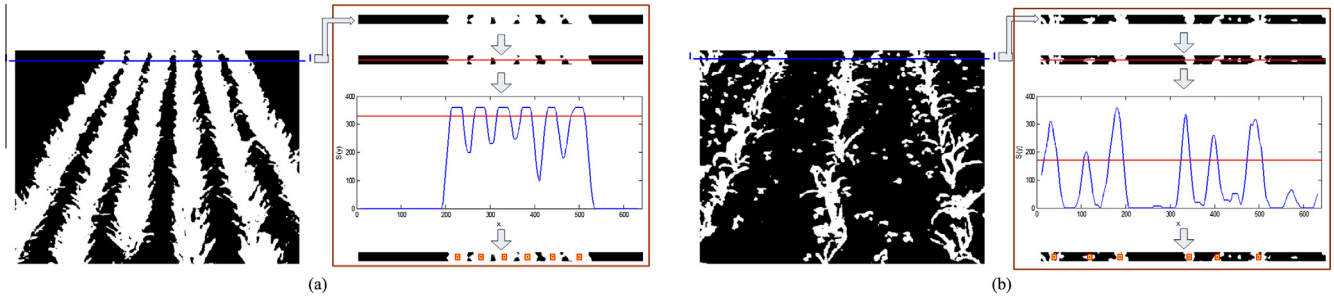


Fig. 6. SROI results on two kinds of images under different weed conditions. (a) Weed-free. (b) Weedy.

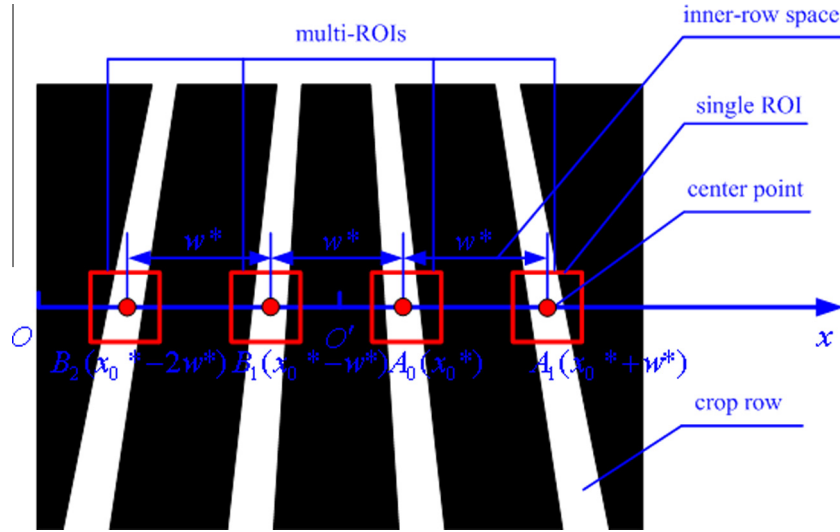


Fig. 7. The schematic drawing that illustrate the estimation of center points of crop rows based on multi-ROIs.

adapt the above same strategy, since the parameters x_0 and w will be changed in a large range during the process of every scan line. In order to speed up the computations, one can use the geometrical properties of the scene (context) to obtain the residual center points. Therefore, one can design a novel approach taking into account of the fact that the corresponding center points of the crop rows varies little in adjacent scan lines. So, it may be feasible to use the values of x_0^* and w^* which have been detected in previous scan line as the initial corresponding values of the current scan line. Then, we give a little disturbance δ ($\delta > 0$) on the basis of x_0^* and w^* of the previous scan line to look for the optimized x_0^* and w^* of the current scan line.

In view of perspective, the inter-row space slightly increases from narrow to wide from top to bottom, the parameter w should also increase during the scanning process from top to bottom. So the current variables of $x_{0_current}$ and $w_current$ can be restricted as follows:

$$\begin{aligned} x_{0_current} &\in [x_{0_previous} - \delta, x_{0_previous} + \delta] \\ w_current &\in [w_previous, w_previous + \delta] \end{aligned}$$

where, $x_{0_previous}$ and $w_previous$ denote the optimal values of x_0^* and w^* in the previous scan line.

According to the Eq. (10), take $x_{0_current}$ and $w_current$ as the initial values of x_0 and w , respectively, we can quickly compute the optimal parameters x_0^* and w^* of the current scan line. Repeat this process until the image is scanned to the bottom.

The result of detection is illustrated in Fig. 9. As expected, all the center points of crop rows are clearly marked.

2.2.4. Real center points confirmation

The above method can be worked well under ideal condition. In practice, because of the effect of perspective, the crop rows size in upper parts are slightly smaller than that of the lower parts in the agronomic image. Moreover, it should be pointed out that the influence will be aggravated with the decrease of the pitch angle of the camera. As a result, in some cases, some pseudo center points, as shown in Fig. 10(a) and (b) were detected in the upper parts of the image.

In order to eliminate these pseudo center points, one can use the classification method. It must be noted that the range of k (see Eq. (10)) decides the number of crop rows. In theory, a specific value of k , corresponding to a specific crop row. So, we can make a classification of center points (including the real center points and pseudo center points) according to the value of k . Now the issue is how we judge the center points belong to a certain value of k can eventually form the real crop row. The simplest method is to accumulate the number of center points (here, the center point must meet the conditions: in a relatively small neighborhood, the number of target points must reach a certain threshold T_1) for a certain k . As a result, if the sum of center points for a certain k less than a threshold T_2 , these center points will be regarded as pseudo center points and will be discarded. Otherwise, be reserved. As can be seen from Fig. 11, the pseudo center points in the upper half of image were removed.

2.2.5. Crop rows detection

After the real center points indicating the centers of the crop rows are determined, the next step is to fit corresponding straight line equation to each crop row.

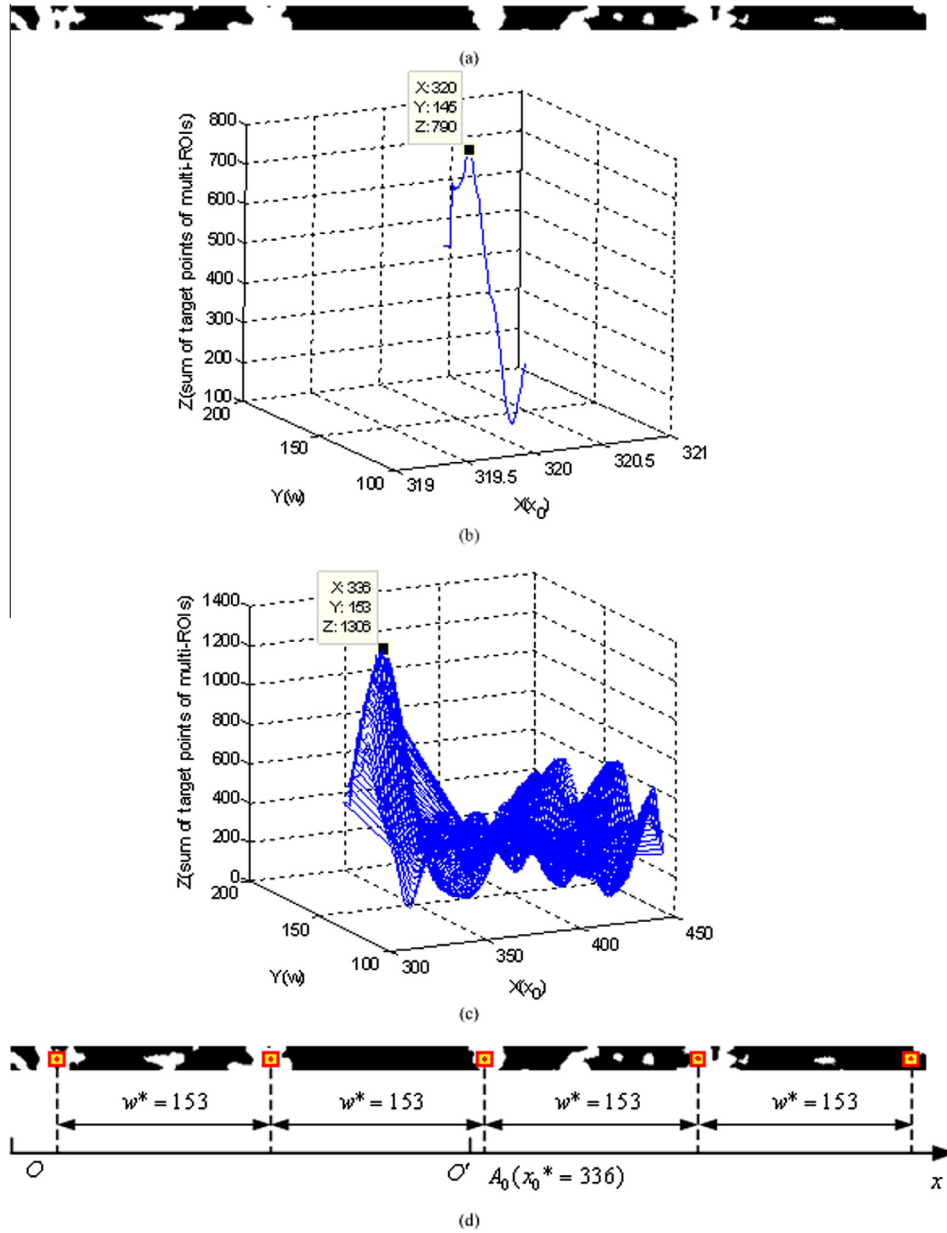


Fig. 8. The acquisition process of the global optimal x_0^* and w^* . (a) The same image strip as Fig. 6(b), (b) 3D diagram of the target points sum of the multi-ROIs under the condition that the w was changed while the x_0 kept unchanged (320), note that the sum reaches the local maximum value when $w = 145$; (c) 3D diagram of the target points sum of the multi-ROIs under the condition that x_0 and w are changed within their own corresponding value range. Note that the Eq. (11) reaches the global maximum value when $x_0^* = 336$ and $w^* = 153$.

Least square fitting is a frequently used method of processing experimental data. It is often to be used to find the best-fitting curve to a given set of points by minimizing the sum of the squares of the offsets of the points from the curve. The sum of the squares of the offsets is used instead of the offset absolute values because this allows the residuals to be treated as a continuous differentiable quantity. Because the functions to be built in our approach are always straight lines, we used the linear least squares method (LSM).

Suppose there are n feature points $((x_1, y_1), (x_2, y_2), \dots, (x_n, y_n))$ in image space, the prediction is given by the following equation:

$$y = \beta_0 + \beta_1 x \quad (12)$$

where, β_0 and β_1 are the intercept and slope of the regression line. The least squares fitting minimize the expression:

$$\varepsilon = \sum_{i=1}^n [y_i - (\beta_0 + \beta_1 x_i)]^2 \quad (13)$$

The least square estimates of β_0 and β_1 can be obtained as follows by solving normal equations:

$$\begin{cases} \beta_0 = \bar{y} - \beta_1 \bar{x} \\ \beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \end{cases} \quad (14)$$

where, $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$, $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$.

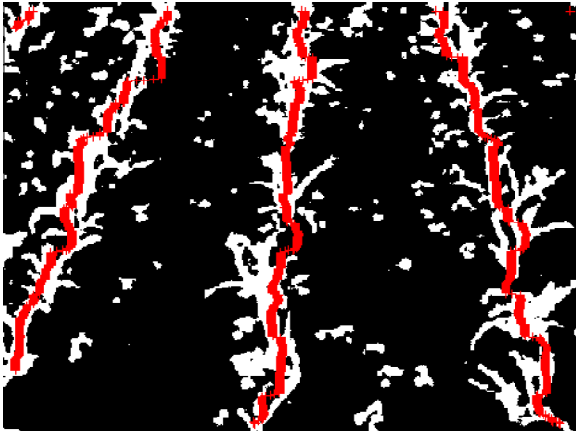


Fig. 9. Estimation of points indicating the center lines of the crop rows ($\delta = 2$).

According to Eqs. (12) and (14), we can compute the intercept and slope of the regression line based on the total center points within a corresponding class, that is to say, the crop rows can be

detected with the LSM. As can be seen from Fig. 12, all of the crop rows were extracted.

3. Results

In order to test the adaptability of the proposed algorithm for different crop under different situations, More than 300 agronomic images were processed in this study. The following is a list of representative images selected from them. The selected images cover the wide range of situations, such as:

- (1) Different distribution of weeds: Fig. 13(a) was taken on June 7, 2008 with scattered distribution of weeds, Fig. 13(b) was captured on May 31, 2008 with spatial aggregation of weeds;
- (2) Different backgrounds: Fig. 13(c) was taken on November 29, 2010 with compact soil, Fig. 13(d) was taken on November 10, 2010 with soft soil;
- (3) Different illumination intensities: Fig. 13(e) was taken on May 31, 2008 with a sunny day, Fig. 13(f) was taken on June 7, 2008 with a cloudy day;

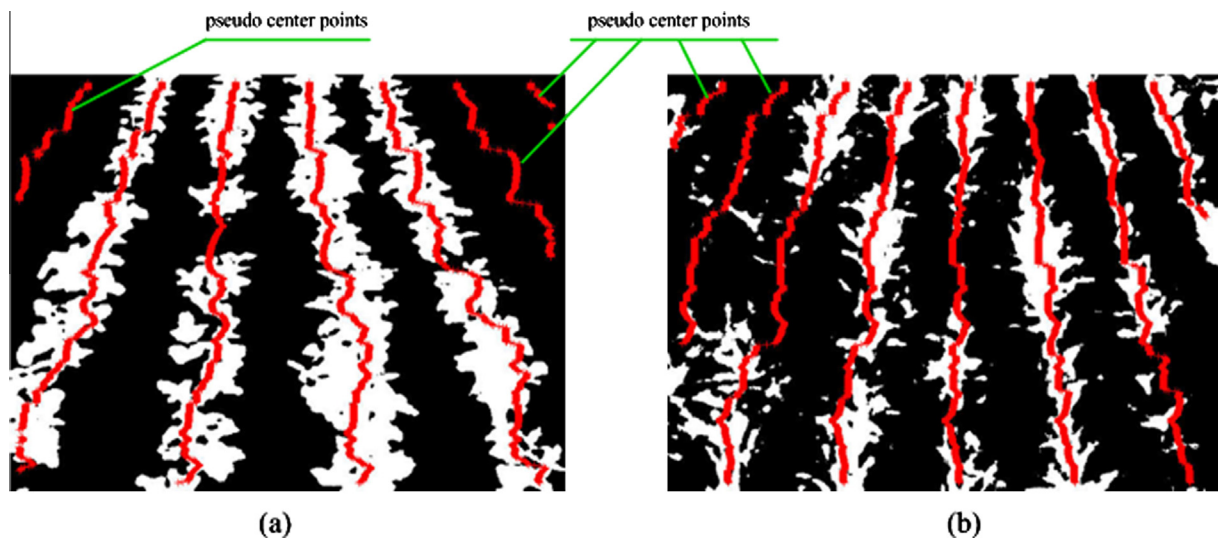


Fig. 10. The result of center points detection with pseudo center points.

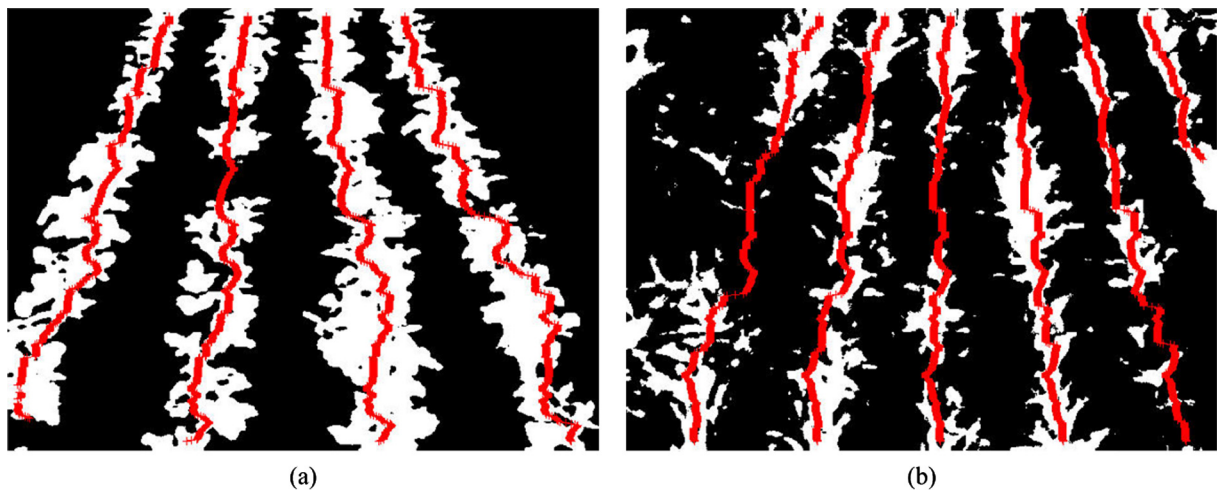


Fig. 11. The result of real center points detection from images in Fig. 10(a) and (b). Here, the size of neighborhood region of center points is 9×9 , $T_1 = 40$ and $T_2 = 50$.



Fig. 12. The result of crop rows detection based on LSM for Fig. 10(a) and (b).

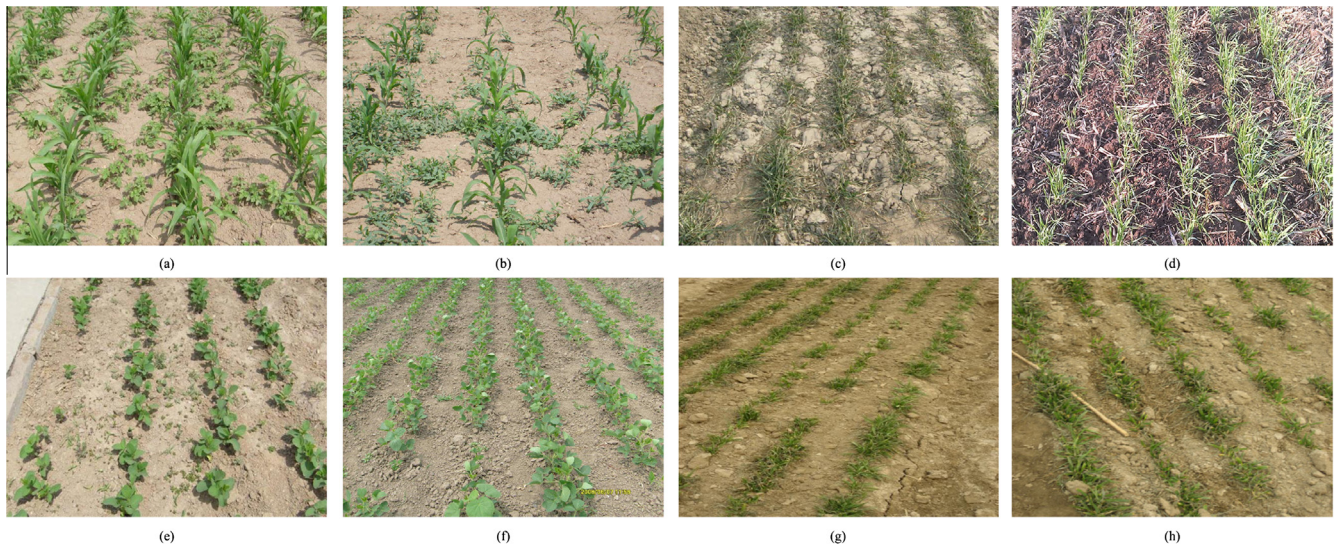


Fig. 13. Agronomic images under different conditions.

- (4) Different camera yaw: Fig. 13(g) and (h) were taken on March 8, 2010 under different angle of yaw.

From the above images displayed in Fig. 3, we can see that after the emphasizing the green value, the green plants including crop and weeds, are identified from the original color images in the corresponding images in Fig. 2(b).

The gray-level images in Fig. 3 were segmented with an automatic thresholding method (Otsu), after which, obtaining a binary image (Fig. 4), where the living plant material (crop or weeds) appears as white pixels and the rest (i.e. soil, stones, shadow) is black. As expected, the self-adaptation threshold algorithm was not affected by the light intensity and the soil background. The green plant was segmented entirely from background.

Fig. 14 displays the confirmation results of center points from images in Fig. 13, which experiences a series of image processing as mentioned above: gray-level transformation, binarization, center points estimation and center points confirmation.

As Fig. 7 shows, the multi-ROIs integrated several ROIs with the same size. So the size of single ROI in multi-ROIs directly influences the velocity and precision of the approach. Practical experience has shown that the dimension of each ROI is 19×19 is suitable when

working with a camera pitch angle of about 40° – 50° . Extensive tests also showed that the method achieves good results over a wide variety of conditions. Tested on different crop (wheat, corn and soybean) and during different years, the center points can be recognized successfully, even when presented with high weed pressure (Fig. 13(a) and (b)). Moreover, it should be pointed out particularly that the method is able to deal with incomplete row structures due to missing plant (Fig. 13(e)) and can automatic count the number of crop rows. From Fig. 13(g) and (h), it can be observed that the algorithm can also achieve a good result even under a condition with a small yaw of camera. The main reason for all of this can be attributed to the method based on multi-ROIs which fuse information coming from several crop rows rather than some single crop row. This strategy seeks a global optimum rather than a local optimal. For a full description and some more interesting examples can refer to Section 2.2.3.

Fig. 15 shows the final result of crop rows detection based on least square method. In LSM, because squares of the offsets are used, outlying points usually have a disproportionate effect on the fit, which may bring an undesirable result. But in our algorithm, during the process of estimation of center points representing crop row, due to the center points obtaining on the current scan

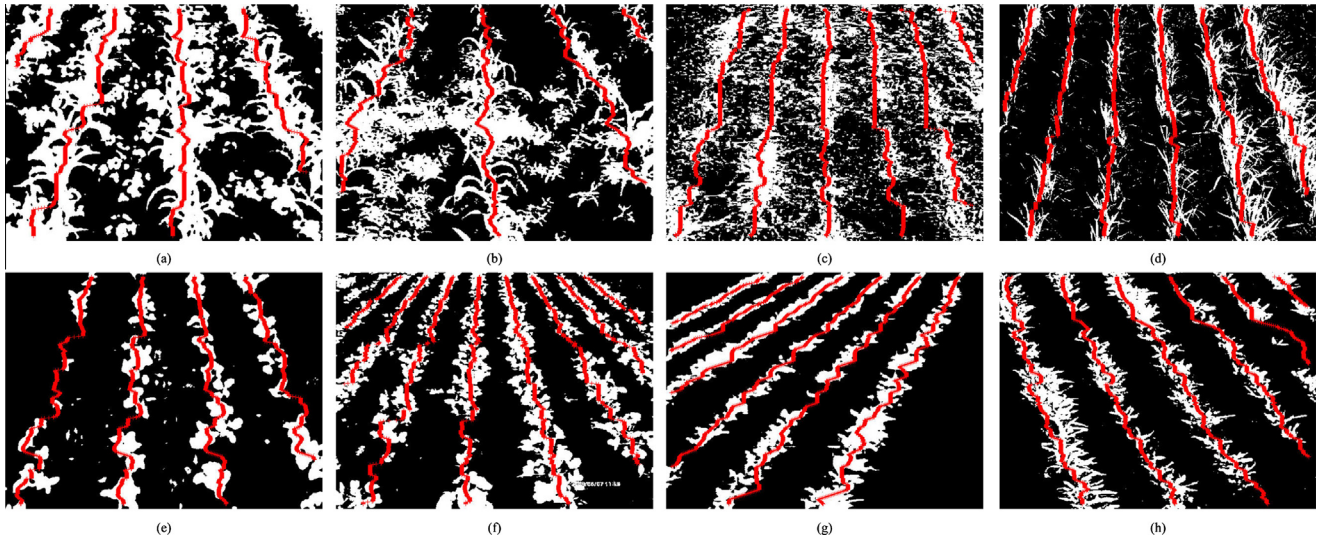


Fig. 14. Result of center points detection. Here, the size of each ROI is 19×19 and $\delta = 2$.

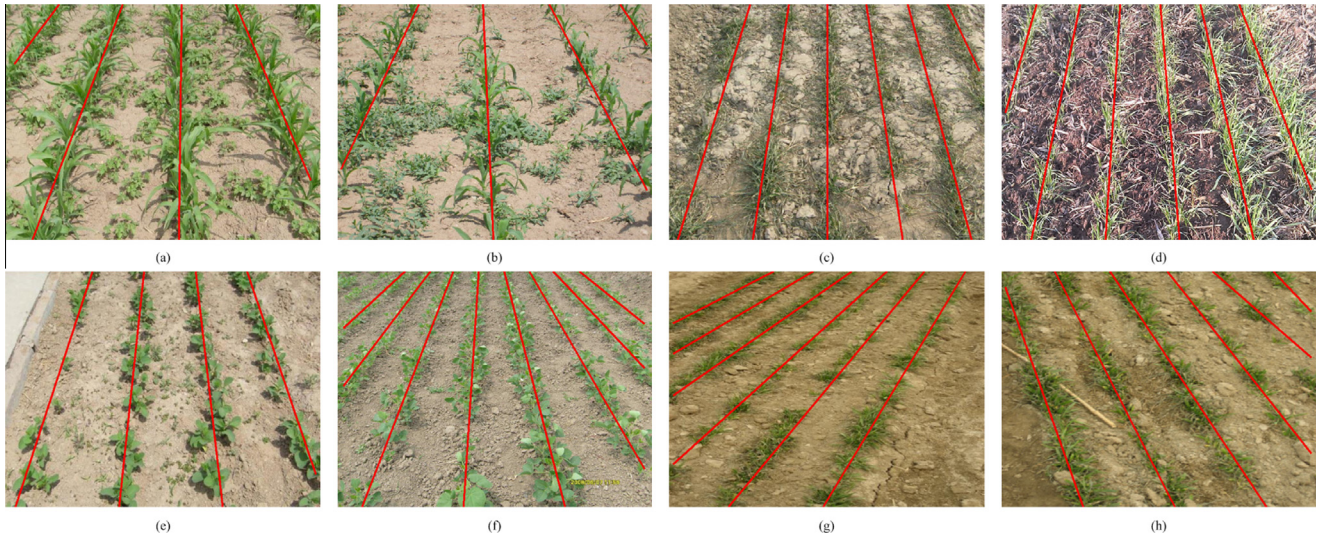


Fig. 15. Crop rows detection based on LSM.

line are based on the center points of the previous scan line, just a very small change, the outlying points does not exist at all. As a result, the applying of LSM achieves the result as we expected.

As mentioned previously, the Hough transform is one of commonly used algorithms for detecting lines in raster images. So we compare the performance of the proposed method with the standard Hough transform. In the first two steps, two algorithms use the same operation, i.e., they have experienced the gray-level and binarization transformation. Then in the following steps, the former used the center points detection based on left–right of crop row edges (Zhang, Chen, & Zhang, 2008) and SHT. The proposed algorithm used the global strategy based on multi-ROIs and LSM (Guoquan, Xing, Shangfeng, Man, & Jiao, 2009).

We tested the performance according to the detection rate, detection accuracy and processing time for different image resolutions. The detection rate, expressed as a percentage, is the ratio between the number of images correctly detected and the number of images determined by an expert human; Detection accuracy aims to measure the deviation between the detected line and the

drawn line; The processing time refers to the average time for the whole test images.

However, the measuring of the approach's accuracy, i.e., detection accuracy, is not quite straight-forward because it is difficult to get true position and direction for the center lines of crop rows due to natural variations in the crop growth stage. Here, we present a simple method to make a rough estimate.

A schematic diagram of the mechanism is given in Fig. 16. In a small strip (denotes the region of crop row), assume that line l_1 is a known straight line and line l_2 is a straight line has been detected. If the angle between l_1 and l_2 is small enough, we think the l_1 and l_2 are coincident (without considering the parallel condition). That is to say, the smaller the angle, the higher the accuracy. So, we can use the angle as a standard against which proposed approach.

The linear equations of l_1 and l_2 are expressed as follows:

$$l_1 : y = k_1x + b_1 \quad (15)$$

$$l_2 : y = k_2x + b_2 \quad (16)$$

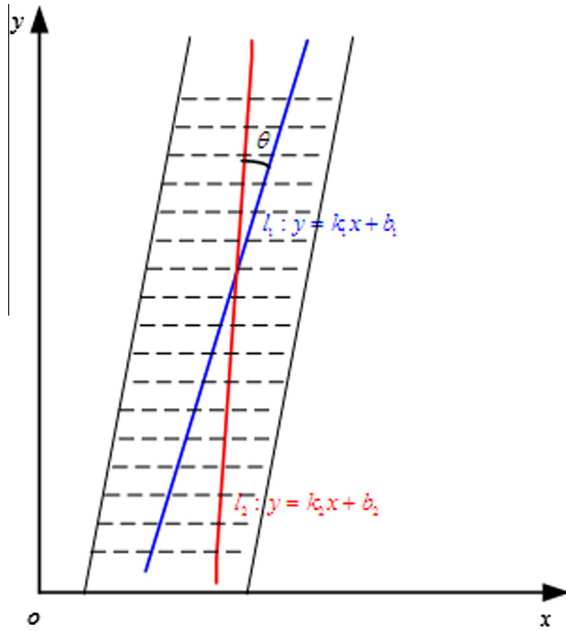


Fig. 16. Schematic diagram of detection accuracy. The blue line (l_1) is drawn by expert human and the red line (l_2) is obtained by detection method using our presented method. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where, k_1 and k_2 are the slopes of l_1 and l_2 respectively, b_1 and b_2 are the y-intercepts of l_1 and l_2 respectively. Let θ be the angle between l_1 and l_2 , then:

$$\theta = \arctan \left(\frac{|k_2 - k_1|}{1 + k_1 k_2} \right) \quad (17)$$

In Fig. 17 we show the results of obtaining the crop rows with two different methods. Note that each angle between the drawn line and the detection line is very small. Suppose there are n lines,

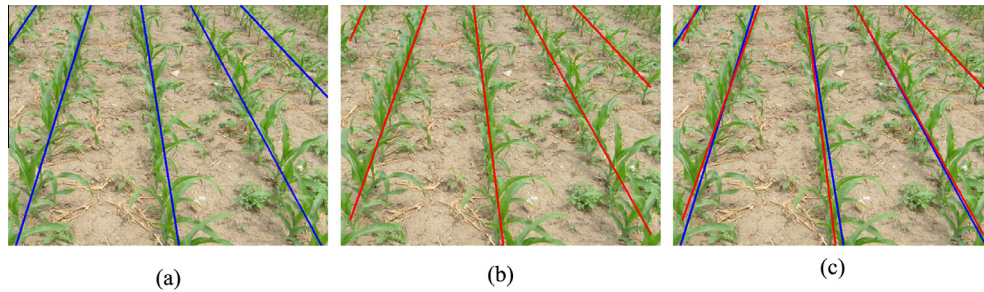


Fig. 17. Comparison of the detected crop rows and the drawn rows. (a) shows the results obtained by hand, (b) shows the results obtained using the proposed method, (c) shows the results obtained using the methods of (a) and (b). The blue lines were drawn by hand, the red lines were obtained by our approach. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Performances of SHT and the proposed method according to the detection rate, detection accuracy and processing time over different resolution images.

Image resolution (pixels)	Detection rate (%)				Detection accuracy (°)				Processing time (ms)			
	SHT		Proposed		SHT		Proposed		SHT		Proposed	
	θ_1	θ_2	ROI1	ROI2	θ_1	θ_2	ROI1	ROI2	θ_1	θ_2	ROI1	ROI2
1024 × 768	91	89	94	91	0.0283	0.0373	0.0012	0.0025	468	296	125	62
800 × 600	90	89	93	90	0.0356	0.0459	0.0028	0.0051	328	226	96	47
640 × 480	88	87	93	89	0.0621	0.0798	0.0063	0.0106	203	125	61	30
400 × 300	86	83	90	88	0.1126	0.1317	0.0769	0.0901	93	62	31	15

Notes: θ_1 and θ_2 represent the quantization precision of angle in parameter space, which are 0.5° and 1° respectively; ROI1 and ROI2 refer the size of ROI in proposed method, which are 19×19 and 9×9 pixels, respectively.

the obtained angles from left to right are $\theta_1, \theta_2, \theta_3, \dots$, respectively. The average of the angles can be computed according to the Eq. (18).

$$\mu = \frac{1}{n} \sum_{i=1}^n \theta_i \quad (i = 1, 2, \dots, n) \quad (18)$$

where, n denote the number of crop rows.

In this study, to get a more convincing results, we test 60 images and obtain the average angle is 0.051° . Obviously, the accuracy is encouraging.

In order to test the effect of image resolution on the detection results, four kinds of different resolution images used in the experiments, which were 1024×768 pixels, 800×600 pixels, 640×480 pixels and 400×300 pixels, respectively. In particular, to accurately test the real-time performance, the two algorithms were coded with C programming language and run under the Microsoft Visual Studio 2008 platform. The test results can be seen Table 1.

In Table 1, from the view of detection rate and detection accuracy, the results of presented method are superior to SHT. That is because the proposed method fuses information from multi-ROIs, which result in an increase in the robustness of the algorithm. Furthermore, the larger size of the ROI, contains more information, so, the detection rate and detection accuracy of ROI1 is higher than ROI2. For SHT, the crop rows detection are recognized by detecting the peaks in Hough space. However, detecting peaks robustly by avoiding spurious peaks is very difficult, particularly in complicated scenes (Furukawa and Shinagawa, 2003). Therefore, SHT is vulnerable to false-positives when handling images with high weed pressure. As for the Hough space quantization precision, when the step of angle was set 0.5° , it produced better detection rate, because it was more specialized Hough space.

In view of processing time, the method put forward outperforms SHT. Two things lie behind the superiority of the processing time for proposed method. On the one hand, the strategies of center points detection based on context information, or, more accurately, the introduced variable σ reduces the time complexity

of the algorithm. On the other hand, it is due to application of LSM. Particularly mentioned, when the size of ROI is 19×19 , the processing time is about 61 ms, it is able to meet the needs of the real-time farmland operation.

In short, according to Table 1, it is clear that the proposed method has a superior performance over SHT either in detection rate, detection accuracy or in average running time.

4. Conclusions

In this paper, an approach of robust recognition of crop rows based on multi-ROIs is presented. Firstly a original color image is transformed into a gray scale image. Then a binarization process is executed based on Otsu's method, which provided the foundation for identifying green plants (crops and weeds). Considering the fact that the spacing between crop rows is equal, a model based on multi-ROIs, making an important contribution for estimating of points indicating the center lines of crop rows. A simple classification method was applied to confirm the real center points indicating crop rows. Finally, the linear least square method is employed for calculating equations of straight lines corresponding to the crop rows.

The proposed method can detect crop rows with all kinds of crop cultivated with rows even in the areas of the field where patches of weed or bare patches occur. In addition, the method has a superior performance over SHT.

However, the research on automatic crop rows detection has still a long way to go considering the complexity of the problem as a whole. Future work may be conducted in regard to three facets. First, the color indices we used only has to highlight the green plants, but the effectiveness drops when the green spectral component becomes less important, thus the proposed approach should be extended to deal with other conditions. Second, the system should be designed with a more intelligent expert, so that combination of all categorized algorithms would be appreciable, neural network for plant and weed classification, machine vision for crop rows detection, fuzzy systems for obstacle avoidance and other techniques would be necessary (Mousazadeh, 2013). Lastly, the fusion of vision sensor with other navigation sensors, such as Global Positioning System (GPS) and inertial sensor, to provide additional information to help with tasks such as end-of-row detection should be an active research topic in the future.

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